

Question 1 :

SR No.	Error	Line Number	Error Type	Error Correction
1	Missing semicolon after variable declaration: int department = 1	11	Syntax Error	Add semicolon: int department = 1;
2	Assigning string literal to integer variable: int marks = "90"	13	Type Error	Use integer value: int marks = 90;
3	Variable score is not declared before use	29	Compilation Error	Declare score variable before printf or replace with a declared variable
4	Assignment used instead of comparison in if(x = 10)	35	Logical Error	Use comparison operator: if(x == 10)
5	Local variable totalStudents shadows global variable	42	Scope Error	Rename local variable or use global variable intentionally
6	Function add() is called but not declared/defined	54	Compilation Error	Declare and define add(int, int) function before use
7	Variable temp is used outside its block scope	60	Scope Error	Declare temp outside the block if required later
8	Array index out of bounds: arr[10] for arr[5]	64	Runtime Error	Use valid index from 0 to 4
9	Printing float variable with %d format specifier	68	Format Error	Use %f for float: printf("%f", salary);
10	Inner count variable shadows outer count variable	74	Scope Error	Rename inner variable or avoid duplicate declaration
11	Variable result is used outside for-loop block	84	Scope Error	Declare result before loop if needed outside

SR No.	Error	Line Number	Error Type	Error Correction
12	if and else statements print the same message	91	Logical Error	Change else statement message according to condition
13	Function displayData() requires one argument but called without argument	99	Compilation Error	Pass argument: displayData(value);
14	Semicolon after while condition creates empty loop	119	Logical Error	Remove semicolon after while(z > 0)
15	Character assigned using string literal: char grade = "A"	124	Type Error	Use character literal: char grade = 'A';
16	Printing integer variable with %s format specifier	128	Format Error	Use %d for integer: printf("%d", number);
17	Dereferencing NULL pointer	132	Runtime Error	Assign pointer to valid memory/address before dereferencing
18	Function calculateAverage() is called but not declared/defined	136	Compilation Error	Declare and define calculateAverage() function before use

Question 2 :

SR No.	Error	Line Number	Error Type	Error Correction
1	Assigning string literal to integer variable: int marks = "90";	10	Type Error	Use integer value: int marks = 90;
2	Missing semicolon after printf("Program Started\n")	18	Syntax Error	Add semicolon: printf("Program Started\n");

SR No.	Error	Line Number	Error Type	Error Correction
3	External variable total is declared but not defined	24	Linker Error	Define the variable globally: int total;
4	Function add() is called before its declaration	35	Compilation Error	Add function prototype before main: int add(int, int);
5	Address of register variable is accessed: &number	39	Compilation Error	Remove register keyword or do not use address operator &
6	Assigning string literal to integer variable: static int salary = "50000";	41	Type Error	Use integer value: static int salary = 50000;
7	Variable department is used outside its scope	47	Scope Error	Declare department outside the block
8	Variable temp is used outside its scope	55	Scope Error	Declare temp before the loop if required outside
9	Array index exceeds array size: arr[10] for arr[5]	59	Runtime Error	Use a valid index from 0 to 4
10	Division by zero: a/b where b = 0	64	Runtime Error	Check denominator before division
11	Assignment operator used instead of comparison: if(value = 100)	68	Logical Error	Use comparison operator: if(value == 100)
12	Both if and else blocks print the same message	76	Logical Error	Change the else message according to the condition
13	External variable total is used in update() but not defined	87	Linker Error	Define total globally before using it
14	Character variable initialized with string literal: char grade = "A";	91	Type Error	Use character literal: char grade = 'A';

SR No.	Error	Line Number	Error Type	Error Correction
15	Variable data is used without initialization	112	Undefined Behavior	Initialize variable before use: <code>int data = 0;</code>
16	NULL pointer is dereferenced: <code>*ptr = 100;</code>	116	Runtime Error	Assign a valid memory address before dereferencing
17	External variable <code>companyCode</code> is declared but not defined	120	Linker Error	Define <code>int companyCode;</code> globally
18	Function <code>calculateAverage()</code> is called but not declared or defined	125	Compilation Error	Declare and define <code>calculateAverage()</code> function
19	String literal assigned to integer variable: <code>int employeeId = "EMP101";</code>	130	Type Error	Use character array: <code>char employeeId[] = "EMP101";</code>
20	Extra semicolon after while condition creates an empty loop	133	Logical Error	Remove semicolon: <code>while(counter > 0)</code>
21	Inner variable <code>y</code> shadows outer variable <code>y</code>	141	Scope Error	Rename inner variable to avoid confusion

Question 3 :

SR No.	Error	Line Number	Error Type	Error Correction
1	Missing semicolon after variable declaration: <code>int choice</code>	24	Syntax Error	Add semicolon: <code>int choice;</code>
2	Missing colon after default in switch statement	55	Syntax Error	Use default: instead of default

SR No.	Error	Line Number	Error Type	Error Correction
3	Array index can exceed valid range: index > 5 check is incorrect	82	Logical Error	Use if(index >= 5) before inserting a new student
4	Loop condition allows access beyond array size: i <= totalStudents	94	Runtime Error	Use i < totalStudents
5	Division by zero in return statement: return total / 0;	114	Runtime Error	Divide by totalStudents: return total / totalStudents;
6	totalStudent variable is not declared	143	Compilation Error	Replace with declared variable: totalStudents
7	Function printGrade() is called but not declared or defined	145	Compilation Error	Declare and define printGrade() function
8	Function calculateRank() is called but not declared or defined	147	Compilation Error	Declare and define calculateRank() function
9	Assigning string literal to integer variable: s.rollNo = "101";	154	Type Error	Use integer value: s.rollNo = 101;
10	Assigning string literal to float variable: s.marks = "90";	158	Type Error	Use float value: s.marks = 90;
11	Array index may go out of bounds: students[roll]	174	Runtime Error	Validate roll number before accessing array
12	Assigning NULL to structure variable: students[roll] = NULL;	184	Type Error	Use a valid deletion method or reset structure members
13	Comparing strings using ==: name == students[0].name	193	Logical Error	Use strcmp(name, students[0].name) == 0
14	Float value returned from calculateAverage() stored in integer variable	203	Type Conversion Error	Use float avg; instead of int avg;

SR No.	Error	Line Number	Error Type	Error Correction
15	File opened in read mode but used for writing: fopen("student.txt","r") with fprintf()	223	File Handling Error	Open file in write/append mode: "w" or "a"
16	File pointer fp may be NULL before file operation	223	Runtime Error	Check if(fp == NULL) before using file
17	File opened but file may not exist: fopen("missing.txt","r")	235	File Handling Error	Check file existence or handle NULL pointer
18	Missing function prototypes for printDetails(), updateMarks(), deleteStudent(), searchStudent(), statistics(), showMenu(), saveData(), loadData()	151-239	Compilation Error	Add function declarations(signature) before main()

Question 4 :

Sr No.	Error	Line Number	Error Type	Error Correction
1	Database size check is done after incrementing totalAccounts	86–88	Logical Error	Check if(totalAccounts >= 10) before adding a new account.
2	Loop condition i <= totalAccounts accesses one extra element	98	Logical Error	Replace with for(i = 0; i < totalAccounts; i++).
3	Withdrawal condition is reversed	139	Logical Error	Replace with if(accounts[acc].balance < amount).
4	Variable totalAccount is undeclared	192	Compile-Time Error	Replace totalAccount with totalAccounts.

Sr No.	Error	Line Number	Error Type	Error Correction
5	Function calculateInterest() is not defined	194	Compile-Time Error	Replace with interestCalculator(); or define calculateInterest().
6	String assigned to integer variable a.accNo	203	Type Error	Replace with a.accNo = 1001;.
7	String assigned to float variable a.balance	207	Type Error	Replace with a.balance = 50000;.
8	Structure variable assigned NULL	228	Compile-Time Error	Reset members individually or use a deletion flag instead of NULL.
9	String comparison using ==	237	Logical Error	Replace with if(strcmp(name, accounts[0].name) == 0).
10	Function generateAverage() is not defined	247	Compile-Time Error	Define generateAverage() or remove the call.
11	File opened in read mode and then written using fprintf()	256–258	File Handling Error	Open file using fopen("accounts.txt", "w");.
12	No check for fopen() failure in saveAccounts()	256	Runtime Error	Check if(fp == NULL) before using the file pointer.
13	No check for fopen() failure in loadAccounts()	267–269	Runtime Error	Check if(fp == NULL) before calling fscanf().
14	Array index out of bounds (arr[10])	311	Runtime Error	Use an index between 0 and 4.
15	Dereferencing a NULL pointer	315	Runtime Error	Allocate valid memory or assign a valid address before dereferencing.
16	Division by zero (a/b where b = 0)	320	Runtime Error	Check if(b != 0) before division.

Sr No.	Error	Line Number	Error Type	Error Correction
17	Using account number directly as array index in depositMoney()	123–124	Logical Error	Search for the account using accNo and then update its balance.
18	Using account number directly as array index in withdrawMoney()	139–146	Logical Error	Find the matching account index before accessing the array.
19	Possible division by zero when totalAccounts is 0	183	Runtime Error	Check if(totalAccounts > 0) before calculating the average.
20	Possible array overflow by accessing accounts[totalAccounts] when full	76–84	Runtime Error	Prevent insertion when totalAccounts >= 10.

Question 5 :

Sr No.	Error	Line Number	Error Type	Error Correction
1	Missing semicolon after int choice	24	Compile-Time Error	Write int choice;
2	Missing colon after default	61	Compile-Time Error	Write default:
3	Database size check is done after incrementing totalBooks	82–86	Logical Error	Check if(totalBooks >= 20) before incrementing.
4	Condition totalBooks > 20 allows overflow at index 20	82	Logical Error	Use if(totalBooks >= 20).
5	Loop condition i <= totalBooks accesses one extra element	96	Logical Error	Replace with for(i = 0; i < totalBooks; i++).

Sr No.	Error	Line Number	Error Type	Error Correction
6	displayBooks() does not check whether any book exists	94–110	Logical Error	Check if(totalBooks == 0) before displaying books.
7	Using book ID directly as array index	120	Logical Error	Search for the matching bookId before accessing the array.
8	issueBook() does not validate book ID	120	Runtime Error	Check whether the entered ID exists.
9	returnBook() does not validate book ID	129–130	Runtime Error	Verify that the entered ID exists.
10	generateReport() assumes at least one book exists and accesses books[0]	136–140	Runtime Error	Check if(totalBooks > 0) first.
11	Possible division by zero when totalBooks is 0	164	Runtime Error	Check if(totalBooks > 0) before calculating average.
12	Variable totalBook is undeclared	167	Compile-Time Error	Replace with totalBooks.
13	Function saveLibrary() called without prototype declaration	170	Warning/Compile Issue	Add void saveLibrary(); prototype.
14	Function calculateFine() is not defined	172	Compile-Time Error	Define calculateFine() or remove the call.
15	String assigned to integer variable b.bookId	179	Type Error	Replace with b.bookId = 101;.
16	String assigned to float variable b.price	185	Type Error	Replace with b.price = 500;.
17	String comparison using ==	196	Logical Error	Use strcmp(title, books[0].title) == 0.

Sr No.	Error	Line Number	Error Type	Error Correction
18	Structure variable assigned NULL	208	Compile-Time Error	Reset members individually or use a deletion flag.
19	updateBook() does not validate book ID	216–223	Runtime Error	Verify the ID before updating.
20	Function calculateAveragePrice() is not defined	230	Compile-Time Error	Define calculateAveragePrice() or remove the call.
21	File opened in read mode and then written using fprintf()	239–241	File Handling Error	Open file with fopen("library.txt", "w");.
22	No check for fopen() failure in saveLibrary()	239	Runtime Error	Check if(fp == NULL) before writing.
23	No check for fopen() failure in loadLibrary()	250–252	Runtime Error	Check if(fp == NULL) before reading.
24	Dereferencing a NULL pointer	275	Runtime Error	Allocate memory or assign a valid address before dereferencing.
25	Array index out of bounds (arr[10])	279	Runtime Error	Use an index between 0 and 4.
26	Division by zero (a/b where b = 0)	284	Runtime Error	Check if(b != 0) before division.
27	Buffer overflow in strcpy()	292–293	Runtime Error	Increase array size or copy a shorter string.
28	malloc(5) allocates insufficient memory for 20 integers	303	Runtime Error	Use malloc(20 * sizeof(int));.
29	Writing beyond allocated memory (ptr[i] up to 19)	305–308	Runtime Error	Allocate enough memory or reduce loop limit.

Sr No.	Error	Line Number	Error Type	Error Correction
30	Accessing memory after free(ptr)	312	Runtime Error	Do not use pointer after free().
31	Double free(ptr) call	314	Runtime Error	Free the memory only once.